

Homework 5. Autoregressive Process

You can get up to 10 points for this homework.

Question 1. The file **SouvenirSales** contains monthly sales for a souvenir shop at a beach resort town in Queensland, Australia, between 1995 and 2001.

Back in 2001, the store wanted to use the data to forecast sales for the next 12 months (year 2002). They hired an analyst to generate forecasts. The analyst first partitioned the data into training and test periods, with the test set containing the last 12 months of data (year 2001). She then fit a regression model to sales, using the training period.

For the following tasks use Python and submit one file incorporating all your answers, named as "H5-YourFirstName-YourLastName.ipynb".

- ☐ A Create a .csv file out of the given data and put it online (you will use a link to it in your program).
- ☐ B Plot the data.
- ☐ C Run the ADF test and check if the process is stationary. If not, apply differencing.
- ☐ D Plot the ACF. What can you see?
- ☐ E Plot the PACF. After what lag the coefficients become abruptly nonsignificant?
- ☐ F Separate your series into a train set and a test set.
- ☐ G Make forecasts over the test set. Use the historical mean method, last known value method, and an $AR(p)$ model for an appropriate p . Apply a window of length 2.
- ☐ H Plot your forecasts.
- ☐ I Measure the MSE. Which method is the best?
- ☐ J Reverse differencing if needed, plot the best forecast and calculate the MAE.
- ☐ K Now, compute a forecast for January 2002.